MNC Weather Report

Retrieves weather station observations from EarthRanger and produces daily summary tables and interactive time-series charts for seven meteorological variables.

What it produces

- **1 daily weather summary CSV** — precipitation, temperature, wind speed, wind gusts, soil temperature, relative humidity, and atmospheric pressure aggregated per weather station and date
- **7 line charts (HTML + PNG)** — one per meteorological variable, coloured by weather station, covering the full analysis time range
- **An interactive dashboard** — one plot widget per chart (7 total)

Requirements

- Access to an **EarthRanger** instance with weather station observations recorded under the subject group **ER2ER - From GMMF** for the analysis period (this subject group name is fixed and not user-configurable)

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1. Load the Workflow

In the workflow runner, go to **Workflow Templates** and click **Add Workflow Template**. Paste this repository's URL into the **Github Link** field, then click **Add Template**:

```
https://github.com/wildlife-dynamics/mnc-weather-report.git
```

Once added, it appears in the **Workflow Templates** list as **mnc_weather_report**. Click the card to open the workflow configuration form.

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2. Configure the Workflow

Data Source Connection

Navigate to **Data Sources** and click **Connect**, then select **EarthRanger**. Fill in the connection form:

Field	Description
Data Source Name	A label to identify this connection (e.g. `Mara North Conservancy`)
EarthRanger URL	Your instance URL (e.g. `your-site.pamdas.org`)
EarthRanger Username	Your EarthRanger username
EarthRanger Password	Your EarthRanger password

> Credentials are not validated at setup time. Any authentication errors will appear when the workflow runs.

Click **Connect** to save.

Workflow Details

Field	Description
Workflow Name	A short name to identify this run
Workflow Description	Optional notes (e.g. reporting month or site)

Time Range

Field	Description
Timezone	Select the local timezone (e.g. `Africa/Nairobi UTC+03:00`)
Since	Start date and time — weather observations from this point are fetched
Until	End date and time of the analysis window

Connect to ER

Select the EarthRanger data source configured above from the **Data Source** dropdown (e.g. `Mara North Conservancy`).

Once all sections are filled, click **Submit**.

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3. Run the Workflow

Once submitted, the runner will:

1. Fetch all observations from the subject group **ER2ER - From GMMF** for the analysis period.
2. Extract seven meteorological variables from each observation's JSON detail field: precipitation, surface air temperature, wind speed, wind gusts, soil temperature, relative humidity, and atmospheric pressure.
3. Compute daily aggregates per weather station — summing precipitation and averaging each remaining variable, except wind gusts which takes the daily maximum.
4. Persist the daily summary as `weather_summary_table.csv`.
5. Draw seven interactive line charts (one per variable), coloured by weather station.
6. Convert all seven charts to PNG at 1 280 × 720 px (2× scale).
7. Assemble the dashboard with one plot widget per chart.
8. Save all outputs to the directory specified by `ECOSCOPE_WORKFLOWS_RESULTS`.

Output Files

All outputs are written to `$ECOSCOPE_WORKFLOWS_RESULTS/`.

File	Description
`weather_summary_table.csv`	Daily per-station summary: sum precipitation, mean temperature / wind / humidity / pressure, max gusts
`precipitation_readings_over_time.html` / `.png`	Daily total precipitation (mm) per weather station
`temperature_readings_over_time.html` / `.png`	Daily mean surface air temperature (°C) per weather station
`wind_speed_readings_over_time.html` / `.png`	Daily mean wind speed per weather station
`wind_gusts_readings_over_time.html` / `.png`	Daily maximum wind gusts per weather station
`soil_temperature_readings_over_time.html` / `.png`	Daily mean soil temperature (°C) per weather station
`relative_humidity_readings_over_time.html` / `.png`	Daily mean relative humidity per weather station
`atmospheric_pressure_readings_over_time.html` / `.png`	Daily mean atmospheric pressure per weather station

More Help

- **Technical Guide:** technical_guide/mnc_weather_report_technical_guide.pdf — pipeline internals and task-by-task reference
- **Issues:** github.com/wildlife-dynamics/mnc-weather-report/issues

Development

This workflow's code (`ecoscope-workflows-mnc-weather-report-workflow/`) is generated from `spec.yaml` and `test-cases.yaml`. After editing either file, recompile and commit the generated changes:

```
pixi run --manifest-path pixi.toml --locked bash -c "./dev/recompile.sh --update"
```

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